

Real Clifford Algebra, Split Octonions, and Triality in Signature (4,4)

A reproducible exact and numerical study

Reproducible exact-real implementation

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Abstract

This dissertation gives an exact-real construction of the Clifford algebra $\text{Cl}(4,4)$, its spin group, the two real half-spin modules, split-octonion multiplication, and split-real triality. The full Clifford algebra is realized by eight integral 16 by 16 matrices and is proved to have rank 256. Its even part preserves two rank-eight eigenspaces of the volume element. Direct commutant and intertwiner calculations prove that these spaces are irreducible and inequivalent as real $\text{Spin}(4,4)$ modules.

An eight-dimensional split composition algebra is independently constructed from Zorn data in an orthogonal basis of signature $(4,4)$. Its multiplication is proved unital, alternative, and norm-composing. Passing to the para-product produces a cyclic invariant trilinear form. The associated related-triple Lie algebra is computed from 512 linear constraints on 84 parameters. It has dimension 28; all three projections are isomorphisms, their eight-dimensional representations are irreducible and pairwise inequivalent, and exact outer operators generate a six-element permutation group.

Two pure-Rust SUNDIALS CVODE studies accompany the exact work. A 24-state transport problem evolves the vector and both half-spin states through one noncommuting Lie-algebra path and preserves all four triality invariants. An 18-state real homogeneous spinor cosmology reconstructs a CPL background and is checked against analytic dilution, density, potential, and Friedmann identities. Every result is generated from hash-pinned source, checked by two independent implementations where practical, and reproduced byte-for-byte.

1 1. Scope and claims

The subject is the split real form in eight dimensions. All vector spaces, matrices, fields, and published numerical outputs are real. The principal claims are:

1. $\text{Cl}(4,4)$ is represented faithfully on a real 16-dimensional module.
2. Restriction to the even algebra and to $\text{Spin}(4,4)$ gives two inequivalent real half-spin modules of dimension eight.
3. A split-octonion multiplication table of signature $(4,4)$ follows from an independently defined Zorn model and satisfies the composition law.
4. The vector module and the two half-spin modules are permuted by an outer six-element triality action preserving one trilinear form.
5. The exact tensors can drive reproducible real ODE studies without changing the audited CVODE engine.

The work does not claim a classification of cosmological observations, a microscopic derivation of dark energy, perturbative stability, or a particle generation mechanism. Those questions require additional mathematics and data. Historical notebooks are treated as provenance and motivation, never as proof.

2 2. Real Clifford algebra in split signature

Let V be a real vector space with ordered orthogonal basis

$(\mathbf{e}1, \mathbf{e}2, \mathbf{e}3, \mathbf{e}4, \mathbf{e}5, \mathbf{e}6, \mathbf{e}7, \mathbf{e}8)$

and bilinear form

$$\eta = \text{diag}(1, 1, 1, 1, -1, -1, -1, -1).$$

The Clifford algebra is the unital associative algebra generated by $\mathbf{V}$ with

$$xy + yx = 2\eta(x, y)1.$$

In particular, the first four basis generators square to $+1$ and the last four square to -1 . The normalization factor two is essential; omitting it is inconsistent with the generator squares.

The standard real classification gives

$$\text{Cl}(4, 4) \simeq \text{Mat}_{16}(\mathbb{R}),$$

and

$$\text{Cl}^0(4, 4) \simeq \text{Mat}_8(\mathbb{R}) \oplus \text{Mat}_8(\mathbb{R}).$$

Thus the full algebra has one irreducible real module of dimension 16, while the even algebra has two inequivalent irreducible real modules of dimension eight. This distinction resolves the most important representation-theoretic ambiguity in the historical material.

3 3. Canonical integral matrix construction

Define three real 2 by 2 matrices

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They satisfy

$$P^2 = 1, \quad N^2 = -1, \quad G^2 = 1,$$

and anticommute pairwise. On the fourth tensor power of the real two-space, set

$$\gamma_k^+ = G^{\otimes(k-1)} \otimes P \otimes 1^{\otimes(4-k)},$$

$$\gamma_k^- = G^{\otimes(k-1)} \otimes N \otimes 1^{\otimes(4-k)}, \quad k = 1, 2, 3, 4.$$

The ordered list

$$(\gamma_1^+, \gamma_2^+, \gamma_3^+, \gamma_4^+, \gamma_1^-, \gamma_2^-, \gamma_3^-, \gamma_4^-)$$

consists of integral 16 by 16 matrices and obeys

$$\gamma_a \gamma_b + \gamma_b \gamma_a = 2\eta_{ab} 1_{16}.$$

The 256 ordered monomials

$$\gamma_{i_1} \gamma_{i_2} \cdots \gamma_{i_k}, \quad i_1 < i_2 < \cdots < i_k,$$

including the empty product, have exact matrix rank 256. They therefore span the full matrix algebra and prove faithfulness without relying on a dimension count alone.

4 4. Volume element and half-spin modules

The ordered volume element is

$$\omega = \gamma_1 \gamma_2 \cdots \gamma_8 = G^{\otimes 4}.$$

It satisfies

$$\omega^2 = 1, \quad \omega \gamma_a = -\gamma_a \omega.$$

The exact projectors

$$P_+ = \frac{1 + \omega}{2}, \quad P_- = \frac{1 - \omega}{2}$$

both have rank eight. Odd Clifford elements exchange their images, while even elements preserve them. The spin generators are

$$\Sigma_{ab} = \frac{1}{4}[\gamma_a, \gamma_b] = \frac{1}{2}\gamma_a \gamma_b \quad (a < b).$$

They satisfy

$$[\Sigma_{ab}, \gamma_c] = \eta_{bc} \gamma_a - \eta_{ac} \gamma_b$$

and the exact $\mathfrak{so}(4,4)$ commutation relations. Restricting the 28 generators to the images of P_+ and P_- gives two eight-dimensional modules.

Three independent tests determine their representation type:

- each even-algebra image has rank 64, the full endomorphism dimension of an eight-space;
- the commutant of each spin action has dimension one;
- the space of intertwiners between the two actions has dimension zero.

The first two statements prove irreducibility over the real numbers. The third proves inequivalence for the fixed Lie-algebra labeling.

5 5. Finite-dimensional representation classification

The Lie algebra $\mathfrak{so}(4,4)$ is the split real algebra of Dynkin type D4. Its four fundamental weights may be labeled so that

- `lambda_1` gives the eight-dimensional vector module V ;
- `lambda_2` gives the 28-dimensional adjoint module;
- `lambda_3` and `lambda_4` give the two eight-dimensional half-spin modules S^+ and S^- .

Finite-dimensional irreducible algebraic modules of the simply connected split group are indexed by dominant integral highest weights

$$\lambda = n_1\lambda_1 + n_2\lambda_2 + n_3\lambda_3 + n_4\lambda_4, \quad n_j \in \mathbb{Z}_{\geq 0}.$$

The diagram automorphism group fixes the central weight `lambda_2` and permutes the three outer weights `lambda_1`, `lambda_3`, and `lambda_4`. Consequently triality acts on every highest-weight family by permuting the three outer coefficients. The spacetime-relevant fundamental triple is

$$(V, S_+, S_-).$$

This statement is more precise than saying that the three spaces are simply the same. They are inequivalent modules for a fixed labeling; an outer automorphism changes the labeling and carries one action to another.

6 6. Split-octonion algebra from Zorn data

Write an element as Zorn data

$$X = (a, u, v, b),$$

where a, b are real scalars and u, v are real three-vectors. Define

$$(a, u, v, b)(c, x, w, d) = (ac + u \cdot w, \\ ax + du - v \times w, \\ cv + bw + u \times x, \\ v \cdot x + bd).$$

Conjugation is

$$\overline{(a, u, v, b)} = (b, -u, -v, a),$$

and the norm is

$$n(X) = ab - u \cdot v.$$

The orthogonal coordinates used by the implementation are

$$\begin{aligned} a &= x_0 + x_4, & b &= x_0 - x_4, \\ u &= (x_1, x_2, x_3) + (x_5, x_6, x_7), \\ v &= -(x_1, x_2, x_3) + (x_5, x_6, x_7). \end{aligned}$$

They give

$$n(x) = x_0^2 + x_1^2 + x_2^2 + x_3^2 - x_4^2 - x_5^2 - x_6^2 - x_7^2.$$

Exact symbolic verification proves

$$n(xy) = n(x)n(y),$$

the left and right alternative laws, flexibility, conjugation reversal, and the quadratic identity

$$x\bar{x} = n(x)1.$$

In the chosen basis the ordinary multiplication tensor has 64 nonzero integral entries and rank eight in each tensor mode.

7 7. Para-product and invariant trilinear form

Define the para-product

$$x \star y = \bar{x} \bar{y}.$$

It obeys the same norm-composition law. The trilinear form

$$T(x, y, z) = \langle x \star y, z \rangle$$

is cyclic:

$$T(x, y, z) = T(y, z, x) = T(z, x, y).$$

This cyclic form is the clean real algebraic carrier of triality. It avoids choosing one preferred spinor and avoids mistaking a single isometry for the full three-way symmetry.

8. Related triples and split-real triality

The related-triple Lie algebra is

$$\text{tri}(\mathbb{O}_s) = \{(A, B, C) \in \mathfrak{so}(4, 4)^3 : A(x \star y) = Bx \star y + x \star Cy\}.$$

Each skew endomorphism has 28 parameters, so the initial space has 84 parameters. Evaluating the defining equation on all 64 basis pairs gives 512 scalar linear equations. The exact constraint matrix has rank 56 and nullity

1. Each of the three projections from the nullspace to $\mathfrak{so}(4, 4)$ has rank 28.

The three induced eight-dimensional representations have commutant dimensions

$$(1, 1, 1)$$

and pairwise intertwiner dimensions

$$(0, 0, 0).$$

They are therefore irreducible and pairwise inequivalent. Direct intertwiner calculations identify the second and third projections with the two canonical half-spin modules from Section 4.

9. Outer six-element action

Cyclic permutation of a related triple defines an order-three automorphism:

$$r(A, B, C) = (B, C, A).$$

Conjugation defines a reflection that fixes the vector position and exchanges the spin positions with the required coordinate conjugations. Its exact matrix has order two. The two operators satisfy

$$r^3 = s^2 = 1, \quad srs = r^{-1},$$

and generate six distinct automorphisms. Because the order-three operator permutes three inequivalent modules, it cannot be induced by conjugation inside the connected spin group. This is the outer triality action.

10 10. Clifford and split-octonion compatibility

Left multiplication by the eight split-octonion basis vectors gives chiral blocks. Combining each block with its conjugate block yields eight real 16 by 16 matrices satisfying the same Clifford relations as Section 3. The space of intertwiners between this multiplication-induced representation and the canonical tensor-product representation has dimension one. A primitive exact intertwiner has rank 16.

This calculation establishes compatibility among three independently defined objects:

1. the tensor-product Clifford representation;
2. the Zorn split-octonion multiplication;
3. the related-triple triality representations.

11 11. Numerical study I: 24-state triality transport

Let

$$y(t) = (v(t), s_+(t), s_-(t)) \in V \oplus S_+ \oplus S_-.$$

Choose one time-dependent Lie-algebra path

$$\begin{aligned} X(t) = & \frac{1}{5}E_{01} + \left(\frac{3}{20} + \frac{t}{40}\right)E_{14} \\ & + \left(-\frac{1}{10} + \frac{t}{50}\right)E_{45} + \frac{t(4-t)}{80}E_{26}. \end{aligned}$$

The selected generators do not all commute and include both compact and split directions. The transport equations are

$$\dot{v} = \rho_v(X)v, \quad \dot{s}_+ = \rho_+(X)s_+, \quad \dot{s}_- = \rho_-(X)s_-.$$

All three initial states equal the unit basis vector. The exact identities

$$\langle v, v \rangle, \quad \langle s_+, s_+ \rangle, \quad \langle s_-, s_- \rangle, \quad T(s_+, s_-, v)$$

are invariants of the ODE.

The application uses pure-Rust SUNDIALS 7.8.0 CVODE with BDF, Newton iteration, a dense linear solver, relative tolerance $1\text{e-}11$, componentwise absolute tolerance $1\text{e-}13$, and maximum step 0.02 . Over $0 \leq t \leq 4$, it writes 41 samples. The verified run used 237 steps and 251 right-hand-side calls. The largest measured invariant drift was less than $3.1\text{e-}12$. A second release run produced byte-identical CSV and JSON.

12 12. Numerical study II: real homogeneous spinor cosmology

Use e-fold time $N=\ln(a)$ and the 18-state real vector

$$y = (\tau, \rho, \psi_0, \dots, \psi_{15}),$$

where

$$\tau = H_0(t - t_0), \quad S = \psi^T \psi.$$

The background parameters are

$$\Omega_{m0} = 0.305, \quad \Omega_{r0} = 0.00009, \quad \Omega_{\psi0} = 0.69491,$$

$$w_0 = -0.861, \quad w_a = -0.60, \quad w(a) = w_0 + w_a(1 - a).$$

The reconstructed potential is

$$U(S) = \Omega_{\psi0} S^{1+w_0+w_a} \exp \left[-3w_a \left(1 - S^{-1/3} \right) \right].$$

Choose a verified real skew Clifford matrix J with $J^2 = -1$. The equations are

$$\frac{d\tau}{dN} = \frac{1}{E},$$

$$\frac{d\rho}{dN} = -3[1 + w(e^N)]\rho,$$

$$\frac{d\psi}{dN} = -\frac{3}{2}\psi + \frac{U_{,S}}{E}J\psi,$$

with

$$E^2 = \Omega_{r0}e^{-4N} + \Omega_{m0}e^{-3N} + \rho.$$

Skewness of J gives

$$\frac{dS}{dN} = -3S, \quad S = e^{-3N}.$$

The continuity equation then gives the analytic CPL density, and substitution gives $\mathbf{rho}=\mathbf{U}(S)$. CVODE integrates independently from the present epoch to $N=-4$ and $N=1$, then joins the branches into 1,201 ordered samples. The verified run used 711 steps and 783 right-hand-side calls. Maximum relative errors were

Identity	Maximum relative error
condensate dilution	6.57e-9
analytic density	8.67e-10
potential reconstruction	7.71e-9
Friedmann closure	3.08e-11

Repeated release runs produced byte-identical outputs.

The calculation establishes a homogeneous background realization. It does not establish perturbative stability. The region with $w < -1$ particularly requires an independent analysis of fluctuations and admissible initial data.

13 13. Reproducibility architecture

The repository separates evidence by ownership boundary:

- `wolfram/` contains exact symbolic definitions;
- `artifacts/exact/` contains hash-linked machine-readable tensors;
- `vendor/sundials_rs` is a read-only Git submodule;
- `studies/` contains application-owned ODE behavior;
- `artifacts/` contains canonical numerical results;
- `notebooks/` contains generated Mathematica and Jupyter presentations;
- `provenance/` contains complete commands for each published artifact.

The exact fixtures are generated twice and compared by SHA-256. Independent Python programs reconstruct their defining identities. Each numerical study is run twice in release mode; semantic checkers recompute invariants from CSV and require byte-identical replay. PowerShell and WSL entry points are both tested.

14 14. Scientific limitations and future work

The exact algebraic conclusions are finite computations over integers and rational numbers. The numerical conclusions are consistency results for two specified ODE systems. Neither category supports claims not encoded in its hypotheses.

Future work should address:

1. global group integration and explicit central-character bookkeeping;
2. orbit and stabilizer classification for null as well as non-null triality triples;
3. perturbations of the homogeneous spinor background;
4. data likelihoods with published covariance rather than central values;
5. geometric field equations coupling the split-signature internal structure to a physical spacetime model;
6. independent numerical comparison on additional platforms.

15 15. Conclusion

The project establishes, in one reproducible real framework, the exact relationship among $\mathbf{Cl}(4,4)$, its 16-dimensional module, the two real half-spin modules, split-octonion multiplication, and triality. The representation distinction is explicit: the full Clifford module is irreducible, its restriction to the spin group splits, and the three eight-dimensional fundamental modules are inequivalent until permuted by an outer automorphism.

The two numerical studies demonstrate how these exact tensors can enter application-owned CVODE models without changing the solver. Their invariants, analytic comparisons, hashes, and limitations are all part of the published result.

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